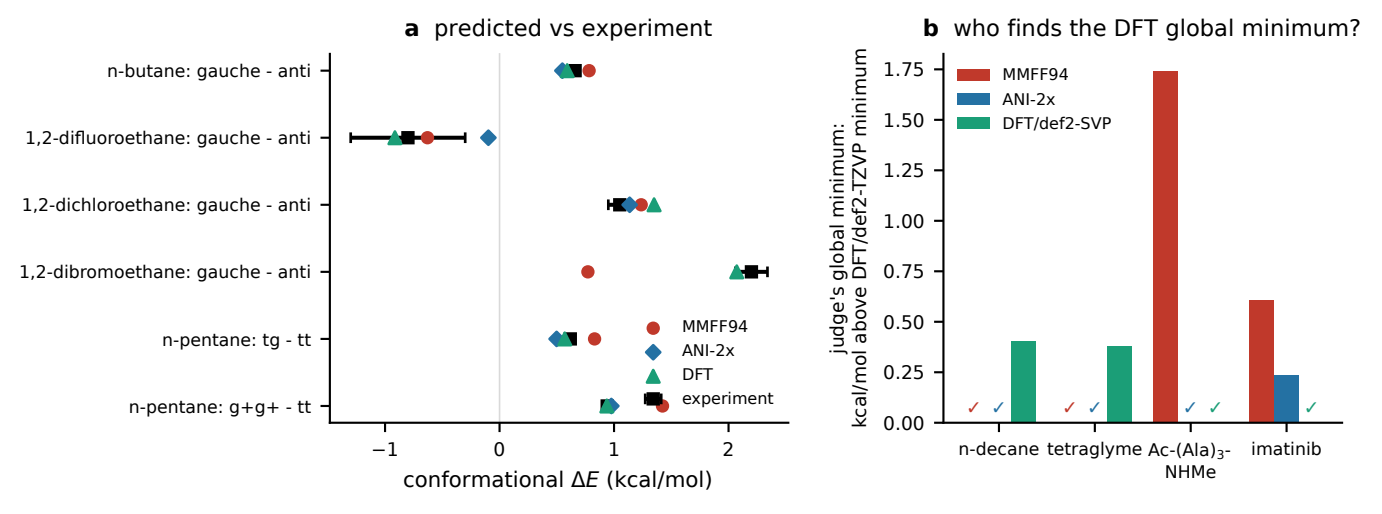
a predicted vs experiment
n-butane: gauche - anti
1,2-difluoroethane: gauche - anti
1,2-dichloroethane: gauche - anti
1,2-dibromoethane: gauche - anti
n-pentane: tg - tt
n-pentane: g+g+ - tt
MMFF94
ANI-2x
DFT
experiment
−1
0
1
2
conformational ΔE (kcal/mol)
b who finds the DFT global minimum?
judge's global minimum:
kcal/mol above DFT/def2-TZVP minimum
MMFF94
ANI-2x
DFT/def2-SVP
1.75
1.50
1.25
1.00
0.75
0.50
0.25
0.00
n-decane
tetraglyme
Ac-(Ala)3-NHMe
imatinib